Opportunities for First-Person Visualization for Sports and Physical Activities

Charles Perin, *University of Victoria, Victoria, British Columbia, Canada*
Charles-Olivier Dufresne-Camaro, *University of British Columbia–Okanagan, Kelowna, British Columbia, Canada*
Maxime Cordeil, *The University Of Queensland, Queensland, Australia*
Jo Jo Gong, *Columbia University, New York, New York, United States*
Christophe Hurter, *ENAC, University of Toulouse, Toulouse, France*
Pourang P. Irani, *University of British Columbia–Okanagan, Kelowna, British Columbia, Canada*
Jade Kandel, *University of North Carolina at Chapel Hill, Chapel Hill, North Carolina, United States*
Ang Li, *The University Of Queensland, Australia*
Tica Lin, *Dolby Laboratories, United States*
Ben Pearman, *University of Calgary, Calgary, Alberta, Canada*
Yalong Yang, *Georgia Institute of Technology, Atlanta, Georgia, United States*
Lijie Yao, *Xi'an Jiaotong-Liverpool University, Suzhou, Jiangsu, China*
Wesley Willett, *University of Calgary, Calgary, Alberta, Canada*

Abstract—New sensing and display hardware is making it increasingly easy to incorporate live data-driven feedback and visualizations to support participants in sports and other physical activities. For example, a jogger wearing a smartwatch equipped with health-related sensors or a pair of smart glasses can monitor their pace and heart rate in-situ and adjust their behavior to meet their goals. However, the diverse contexts in which physical activities take place makes designing for such experiences very different from designing traditional desktop-based data visualizations. Activities may occur in dynamic environments (outdoors in the rain vs. indoors at a gym), involve different social dynamics (volleyball vs. solo hiking), and afford different input and output modalities (encumbered hands while rock climbing vs. jogging). Building upon an IEEE VIS workshop that brought together experts in sports, augmented reality, and situated visualizations, we identify five areas and 21 opportunities for visualization and interaction research for first-person sports and physical activities.

First-person visualizations for sports and physical activities present information relevant to the participant in-situ and in real-time. Unlike more analytic sports visualizations, which seek to uncover trends and optimize players' and teams' performance over longer periods, first-person visualizations present information to participants immediately within the flow of their activity. Yet, sports and physical activities are rich and varied, spanning a wide range of environments, employing different equipment, and prioritizing diverse biometrics, skills, and goals. Building on recent advances in input and output hardware, first-person visualizations hold the potential to enhance participants' athletic experiences—but also raise a host of new design and technical challenges and opportunities for visualization research.

In the visualization community, sports visualization and visual analytics [1] have gained traction in the past decade [2]. Computer vision systems [3], human-activity recognition systems, and wearables can now

This is the Author's version of the work.

be used to track and classify events and athletes with high accuracy and minimal lag, and novel visualization techniques provide spectators and coaches with real-time access to information during sports events. However, participant-focused, first-person research has been limited, in part due to the technical complexity and multidisciplinary challenges it entails. Early research in this space, such as the exploration of situated mixed-reality visualizations for interval running [4], highlights how systems must account for divided attention—where participants' main focus is on the activity and where lapses in attention can lead to poor performance or injury. The physical environments in which activities are conducted can also introduce perceptual challenges, which limit interactions with the system [5]. Finally, participants using these visualizations have a variety of skills and goals, requiring diverse and highly-flexible designs, even for the same activity [6], [7].

With these factors in mind, we highlight research opportunities for first-person visualization systems for sports and physical activities. These opportunities build upon the outcomes of the IEEE VIS 2024 workshop on *First-Person Visualizations for Outdoor Physical Activities* [8]. In this workshop, two dozen researchers specializing in sports, augmented reality, and situated visualizations proposed and examined a rich set of activities, ranging from individual mindfulness-focused practices like stretching, to team-based sports like volleyball, to hazardous embodied activities like outdoor rock climbing. As a group, we distilled unique characteristics, goals, and challenges that first-person visualization systems should consider. We then reviewed, coded, and condensed observations and artifacts from the workshop into 21 opportunities organized within five areas: 1) *goals and information needs*, 2) *data acquisition and processing*, 3) *perception and cognition*, 4) *interactions*, and 5) *sociality*. We illustrate these opportunities through sidebars showcasing six representative activities (hiking, surfing, weight training, rock climbing, volleyball, and soccer), highlighting directions both for new visualization designs and for broadly-applicable visualization research.

GOALS & INFORMATION NEEDS

Meeting the information needs of participants engaged in a sport or physical activity requires understanding their goals (e.g., fitness, mental wellness, or socializing).



Understanding participants' information needs across a wide range of activities, situations, and individuals. Initial examinations of health data queries of smartwatch users [7] and of live information needs of runners [6] suggest that individuals' goals as well as the information they wish to access during activities are unique and dependent on the situation. *How similar or diverse are these needs? Are there general design patterns and displays that work for a wide range of activities, people, and situations?* Similar questions arise when considering longer-term goals related to fitness, skill, behavior change, mental wellness, and socialization, where goals are varied, situation-dependent, and changing. In addition, participants may struggle to articulate their goals or may not have the cognitive or interactive bandwidth to do so during the flow of an activity. Designing technologies for implicit, seamless, and non-disruptive data capture is complex, because information needs, even for a single athlete, can vary widely over the course of an activity.



Designing mixed-initiative systems that are appropriate to the activity and context. Individuals' needs can often differ even within a single activity, and it can be difficult to predict what information is useful at any given moment. For example, a hiker might focus on fitness, discovery, navigation, and form all over the course of a single trail. Relaxed or casual activities prompt different information needs than intense training situations, where access to high-frequency data with precise feedback may be critical. Performance situations, where participants must focus all their attention on the activity itself, introduce additional cognitive or interactive constraints. Similarly, activities that alternate fast-paced action with breaks, such as volleyball, lend themselves well to systems that collect and provide information during downtimes. Finally, participants likely have different needs in team sports such as volleyball and football based on their position and role (a striker needs different information than a goalkeeper), and in activities like climbing, where the information needs of climbers, belayers, and viewers are very different.



Addressing widespread but unmet needs through multidisciplinary collaborations. Given the heterogeneity of activities, participants, and situations, building a deeper understanding of the space of information needs requires a deeper understanding of their history, practice, ethos, and demographics. This calls for collaborations with other disciplines, for example, to ensure that new feedback designs effectively align with best practices in health and kinesiology research.

DATA ACQUISITION & PROCESSING

Creating visual tools for sports and activities also calls for new datasets, sensors, and data processing pipelines capable of deriving the information drive them.

HIKING – SLOW-PACED IN EXPANSIVE SETTINGS



HIKING is an outdoor activity with **low barriers to entry**, both in terms of skills and equipment, making it **highly accessible** to a wide range of people and fitness levels. Hikers have different (sometimes competing) **goals** including leisure, fitness, well-being, socialization, solitude, discovery, or experiencing nature.

The need for visualizations to support hikers may be fundamentally different than more performance-focused activities, because there is a high risk that any technology becomes an annoyance and interferes with the hiker's enjoyment of their activity. However, we envision three cases where visualizations can promisingly address information needs of hikers. First, navigation is a central concern, especially as the length and complexity of a route increases, when participants are unfamiliar with the terrain, or when weather conditions evolve. While hikers tend to rely on smartphones or smartwatches with pre-loaded maps, **in-situ navigational aids** could be provided, for example on smart glasses or integrated with equipment like hiking poles. Second, hikers often desire new insights about the landscape, topography, fauna, flora and history of the trails, and **educational situated visualizations** could be integrated into hiking activities. Finally, people often hike in small groups for social and safety reasons. While most hikers are unlikely to adopt technologies that disrupt social interactions, new **collaborative decision-making interfaces** might help groups coordinate around stops or choose which path to take next based on energy, difficulty, or scenery. Ultimately, first-person, situated visualizations should deliver information without forcing viewers to disengage from their surroundings or companions, enabling efficient access to data while maintaining presence and social connection.

Collecting large sample corpora of activities. Collecting body sensor metrics, movement data, and environment scans will allow the research community to set ideation and prototype sessions in pseudo-realistic settings while *ex-situ* (for example, in a lab), and analyze critical situations specific to activities. Although activity-specific data is needed to design context-appropriate visualizations, many activities have overlapping needs which can be addressed by similar data (like body pose for weight lifting or surfing).

Developing pipelines to support dynamic data needs. Data is typically pre-collected and pre-processed, before being encoded and rendered in visualizations. This is the case, for example, if a hiking route is pre-planned or a fitness target is set beforehand. In many cases, however, the most relevant data for an activity is unknown in advance. As information needs evolve and performance, weather conditions, or gear change, new data must be sensed, processed, analyzed, and visualized. Furthermore, special considerations must be taken as sensors may be unusable if the battery life, processing power, or connectivity of the hardware prevent them from being used consistently across the full duration of an activity.

Creating novel approaches to extract meaningful environmental data in real time. In relatively constrained environments, such as a climbing crag or a volleyball court, modern technology can be used to create maps and 3D models with which to render first-person visualizations. However, more open and changing environments, such as those encountered when hiking or surfing, are more challenging to model.



SURFING – Dynamic environments requiring intense focus

SURFING is a free-flowing outdoor sport that is **physically demanding**, **highly dynamic** and dependent on **natural conditions** that must match the surfer's skill level. The environment is unstable and changing, shaped by shifting waves, **weather conditions**, other surfers, and hazards. Surfing unfolds in **distinct phases**: bursts of intense paddling and riding are punctuated by **extended downtime** both in the water and on the beach. The embodied and time-sensitive nature of surfing makes feedback on body placement, balance, and movement highly valuable, though difficult to access in real time given its **high attentional demands**.

The main challenge for augmentation lies in **data capture**, as water is a harsh environment for electronics. Video remains the most common recording tool, while emerging **wearable and board-mounted sensors**, like those embedded into wetsuits, promise to capture biometric and kinematic data such as posture, balance, and board dynamics. In parallel, **environmental sensors** and predictive ocean models can deliver fine-grained insights into waves, currents, and hazards. Together, these advances create opportunities for visualization at different stages of surfing. Real-time situated visualizations—for example, lightweight feedback through smart glasses—could provide **immediate guidance during rides**, provided they are adapted to withstand water and strong currents. AR technology may furthermore be highly effective outside the surf zone to support teaching, analysis, and **muscle programming**—rehearsing movements on land to accelerate skill acquisition that would otherwise be limited by the availability of suitable waves.

There, recent advances in AI, particularly in zero-shot classification, can allow for quick and accurate object detection based on free-text queries such as visual input (for example, through cameras on AR glasses), thereby reducing the need for specific training datasets for activities, provided computing resources are available.

Deriving metrics that meet abstract and complex information needs. Many information needs can be impossible to satisfy if they depend on information that systems may not have access to (*How many other people have taken this trail? Will this next wave break or not?*), require complex sense-making (*How much longer should I run along this trail to burn the same amount of calories as last time?*), or which are inherently unknowable (*What move will my opponent make next? Will one more attempt result in an injury?*). Individual sensor types are unlikely to provide sufficient feedback for many information needs. In response, systems will likely need to exploit multiple sensors and combine them to produce richer metrics that are, in aggregate, more relevant to the person's needs. For example, AI-driven systems could provide predictions and estimates of more complex metrics and allow for iterative sensemaking to refine them.

Conveying data uncertainty visually. Data collection and processing can be complicated by the physical nature of activities and the environments in which they take place, as well as the technical limitations of current sensors and processing hardware. For example, for hazardous activities requiring precise and fast movements like surfing or rock-climbing, off-the-shelf posture sensors may not offer the level of precision and resolution necessary to produce useful visualizations. Foot angle when surfing is not useful if imprecise (low resolution), and the position of other football players on the field is not useful if processed too late (high latency). Similarly, climbers who rely on navigation tools on remote walls often face issues



WEIGHT TRAINING – EMPHASIS ON QUANTIFICATION AND SOCIAL SETTINGS

WEIGHT TRAINING focuses on **repeated targeted movements** to achieve strength, flexibility, and endurance goals while minimizing the **risk of injury**. Weight training sessions are often broken up into individual repeated exercises, which can involve different muscle groups, movements, and equipment. Hands are typically occupied and lifters often focus attention on their breathing and form during lifts, both of which add **interaction constraints**. A key challenge in weight training is the **lack of body visibility** which can make it hard to recognize whether exercises are being performed with proper form. Physical mirrors, including smart mirrors, can offer some guidance, but many exercises and equipments limit full-body visibility.

Wearables and smartphones have served as the main visualization platforms for weight training. However, they have limited capacity for input during exercises and **equipment integration** could, for example, provide direct visual feedback on breathing, or allow lifters to indicate their level of comfort during a set. Providing **body-pose feedback** in-situ is a key challenge because poor form considerably increases the risk of injury, especially for novices. Because weight training sessions are punctuated by frequent pauses between sets, it is well suited to **adaptive visualization complexity** approaches that minimize visual detail during efforts and afford more exploration and feedback during downtimes. Hands-free AR displays are well-suited to the activity, but may introduce **social tensions** in gyms or other public settings where others may perceive that they are being recorded.

with inaccurate position sensing and limited access to map-based information. When the data is not perfect, visualization can help make the data at least useful if the relevant uncertainty is communicated effectively.

PERCEPTION & COGNITION

Many activities have high cognitive and attentional demands to perform well and remain safe. Successful first-person visualizations must be designed to be aligned with those demands, while providing useful information or feedback.

Integrating first-person visualizations seamlessly with the activity and context. The visual and cognitive load of an activity can limit the complexity of visualizations that people can manage, and some representations (like renderings of body poses) are inherently more complex to interpret when viewed from a first-person perspective. The limited time participants can afford to disengage from an activity (for example, between movements or "sets") may call for simple glanceable displays [9]. In addition, making visualizations more prominent or attention-grabbing can ensure that participants notice them even when deeply engaged. For example, heart rate and pace alarms on contemporary running watches often use audio, haptic, and visual feedback to draw runners' attention. However, this approach risks distracting participants, especially as visualizations grow more complex, and can pose safety risks in high-focus and high-speed activities such as surfing.

Establishing empirical guidelines for perceptually-effective first-person visualizations. Because first-person visualizations on head-mounted displays can be rendered on top of physical environments, they present compelling opportunities for peripheral and ambient data displays. However, their utility depends heavily on the good integration between visuals and the real-world to preserve the legibility of both. Objects in the



ROCK CLIMBING is **physically demanding** and **highly embodied**, requiring climbers to marshall **intense focus and attention** while undertaking complicated physical movements in sheer environments. Rock climbing is also characterized by its **attention to safety**, given the potentially high consequences of an uncontrolled fall. As a result, climbers spend much of their time physically engaged both with the climbing surface and with safety **equipment** (including harnesses, helmets, ropes, belay devices, and protection gear), leaving little bandwidth for additional input and output. Climbing also tends to be **highly collaborative** with **distinct roles**, with pairs or teams of climbers actively supporting and protecting one another throughout an ascent, often with relatively **limited communication**.

New data sources like 3D models of climbs (captured via photogrammetry and LiDAR) have the potential to facilitate new **situated visualizations of routes, features, and past climbs**, which could improve climbers' ability to plan for ascents and navigate during them. The continuous presence of safety equipment throughout climbs also creates new possibilities for **equipment-integrated visualizations** and feedback seamlessly in and around equipment like ropes, carabiners, or belay devices. Additionally, climbing represents an opportunity to explore new visualization approaches that address the **differing viewpoints and information** needs of climbers, belayers/spotters, and other participants.

environment can interfere with visualization elements, occluding or biasing the perception of their encodings. Yet, classical guidelines from visualization, HCI, and vision science research for on-screen visualizations (whose backgrounds are usually static) are routinely applied to this new fundamentally-different domain—despite a lack of formal validation—leading to usability and user-experience concerns.

Conducting graphical perception, legibility, and depth perception studies to inform new adaptive visualizations. Perceptual challenges arise when viewing visualizations in dynamic environments. Rapid body movements and changes in contrast can render visualizations uninterpretable. Frequent focus-switching between the visualizations and the environment can induce visual strain, especially if the elements lie at different depths. In those contexts, first-person visualizations could be designed to vary in complexity, density, color encodings, scale, and placements in real-time based on viewing conditions, task at hand, and situation. For example, visuals could be introduced only at key decision points and areas of visual focus along a new climbing route.

Blending displays into the activities and their environments. Displays introduce a variety of specific perceptual issues. For example, current wearables like optical-see-through (OST) displays (most smart glasses) and smartwatches suffer from limited resolution, optical distortions, poor color reproduction, and poor performance under bright conditions. Therefore, different display types may be better-suited for specific types of activities and environments. For example, indoor activities with movements that restrict access to wearable displays may be better suited for OST displays.

VOLLEYBALL – TEAM COORDINATION WITH BURSTY PACING



VOLLEYBALL is characterized by the complex and **rapidly-shifting relationships** between the players on the court. In addition to the physical agility and dynamism necessary to move around the court and field the ball, players must anticipate opponents' actions and coordinate their movements with those of their teammates. The fast-paced nature of the sport means that players have **little bandwidth for visual input** and any augmentations designed to help players understand and anticipate others' actions have **low tolerance for latency**. Similarly, visualizations shared by teammates likely need high **spatial and temporal coordination** to avoid confusion or asymmetric interpretations. Information asymmetry between teams can further lead to concerns about **fairness**, especially if access to that information conveys a competitive advantage.

Given these constraints, volleyball may benefit from visualizations that support enhanced **awareness and coordination during training** as opposed to during live play. It may also hold opportunities for designs that take advantage of the natural **pauses in play**, allowing players to **reflect on past rallies** and **plan for upcoming ones**. More generally, the defined volume of a volleyball court also makes it an intriguing testing ground for **spatially-anchored** visualizations that might help players identify patterns in their own movement, coordinate strategies with their teammates, and better understand their opponents' play.

Foldable, textile-based, and other freeform displays can better fit already existing equipment and practice.

Crafting tailored ideation and prototyping tools. While many mixed-reality visualization frameworks have been released, there is a notable lack of tools that can faithfully simulate real-world environments or viewing conditions on OST displays. Without tools capable of simulating the unique conditions, environments, and contexts in which activities take place, researchers and designers cannot streamline perceptual studies, design explorations, and evaluations of first-person visualizations. They must deploy and test prototypes directly in real-world environments, which requires access to complex locations, conditions, or gear, is cumbersome and time-consuming, and sometimes just impractical, such as with surfing.

INTERACTION

Because of their varying levels of intensity, gear, environments, and other factors, visual tools for sports and activities may vary radically in terms of the kinds of interactions they support, and also call for different interaction design approaches.

Engaging with participants to co-design, deploy and evaluate human-in-the-loop sensing systems. Participants may ask questions from the system (*How does my current performance compare to my previous attempts?*), or they may directly analyze the data shown (*What has my pace been this session?*). However, activities often already involve extensive equipment, which can limit one's ability to interact with the system. For example, thick gloves may block tactile inputs on commercially-available wearables, and ski poles may constrain the range of motion and accuracy of tactile or gesture-based interactions. Activities them-

FOOTBALL – FLOWY AND CONTINUOUS WITH PHYSICAL CONTACT



FOOTBALL (or Soccer) is a team sport played over a **vast field** that requires **high physical and tactical abilities**. Two teams of up to 11 players share the same pitch, with players dynamically alternating between offensive and defensive roles. **Physical contact between players** is frequent, and continuous play can occur over long periods, often without breaks. Players can also move freely on the pitch, which affords **dynamic and adaptive strategies** based on game events.

Given the continuous nature of the sport, first-person visualizations could be leveraged during breaks (throw-ins, substitutions, and between periods) to **adjust tactics**. During real-time play, visualizations might support **split-second decisions**, informed by data. Accessing **historical data** in real-time could help players opt for plays that maximize their chances of scoring or defending, though at the risk of introducing **competitive advantages**. Beyond individual decisions, football requires high coordination between teammates for success, such that any **visualization feedback must be consistent with other players** to avoid confusion.

selves may further constrain the input interaction space through body movements (uninterrupted sequences, postures, balance) or the environments in which they are performed.

Designing new activity-focused wearables that integrate into existing equipment and gear. Sports and physical activities have evolved over time to integrate gear, equipment, and technology. Instead of adding more technology and devices that might disturb the ecosystem that makes an activity pleasurable, we envision integrating sensors into sports equipment rather than adding more standalone devices. For example, ski poles or free weights could support tactile input and afford haptic feedback to facilitate interactions mid-activity, and e-textile garments could incorporate tactile surfaces to minimize reliance on smartwatches or smartphones for input interactions.

Leveraging activity-specific non-traditional input modalities. New lightweight input modalities may help address some of the accessibility and attention constraints of current systems, be more situation-appropriate, and compatible with the practice of the activity. For example, subvocalization or mouth-based interactions can afford short and discrete input interactions that may generalize well across activity types. However, speech may not be a viable input channel in quiet spaces (for activities like weight-training), in noisy outdoor settings (like surfing), or for social activities (like football). Physically-demanding activities like sprinting or lifting may also make speech uncomfortable or even impossible. Brain-computer interfaces might prove useful for lightweight interaction during activities, particularly in cases where their inputs are integrated with other contextual and environmental data streams.

Designing interfaces that are compatible with, and will withstand the constraints of activities. The most appropriate and useful interaction modalities for an activity may require computing resources that are not readily available on wearable technology or usable over long periods of time (such as during a hike). For example, input techniques that rely on cloud computing may not be practical for activities conducted in remote areas with limited connectivity, and water-sensitive hardware would not be appropriate to use when surfing. Addressing these challenges requires collaborations with experts in ergonomics, hardware, and material sciences.

SOCIALITY

Like nearly all ubiquitous computing systems, new tools for sports and activities have social implications. While collaborative or social features may bring new opportunities for team-based and communal activities, new research must also navigate potential conflicts with existing norms and values.

Integrating existing trends in social technologies for live activities. Social technology is extensively used in sports. Activity-specific social networks and services (including Strava for runners and cyclists, Mountain Project for climbers, and AllTrails for hikers) intermediate relationships between enthusiasts by allowing them to share routes or workouts and participate in virtual competitions. However, accessing these platforms in-situ (for example, consulting climbing route details while on the wall or accessing live rankings of Strava segments) is poorly supported by existing devices. First-person visualizations provide new opportunities to create, situate, and share information about practices, equipment, performances, and more with friends, followers, and competitors.

Considering the practice, etiquette, ethos and aesthetic of activities. If technology offers promising ways to transcend the experience of a sports or physical activity, social (and in some cases legal and ethical) tensions often surround the introduction of new technologies to existing activities. Even for non-competitive activities, introducing new equipment or augmentations can raise questions about their appropriateness and the extent to which they change the activity itself. For example, tensions around the use of older technologies like drills and even commercial guidebooks have long created divides within climbing communities, where they can be seen either as enabling tools or as shortcuts that clash with the ethos of the sport. Similarly, capturing data and visualizing information in-situ using devices like smart glasses can introduce privacy challenges and raise concerns about surveillance.

Characterizing the social and competitive impact of asymmetric information and visualizations. Competitive sports and activities are often deeply concerned with the extent to which new technologies offer a competitive advantage and impact accessibility to new participants. Whereas in most activities today, participants generally see the same things, first-person visualizations create the potential for bigger information asymmetry between participants. This divide has important implications both for perceptions of fairness, accessibility, and the extent to which participants are truly engaging in the same activity. Interested participants could notably be "priced out" from organized activities if they require new expensive technologies. Here, visualization researchers may take some cues from video games [10], particularly massively multiplayer online games, where both opportunities for virtual sociality and experiences of information asymmetry are common.

Envisioning new technology-integrated variations of sports. The use of visualizations in sports can take different philosophical paths, with opportunities for these tools to support existing activities and modes of competition or to fundamentally alter them. Augmentations designed to support learning, training, goal-setting, and safety have the potential to subtly augment activities. Alternatively, data, analytics, and visualizations offering large competitive advantages to participants who use them have the potential to lead to wholesale evolutions of activities, and even spawn new ones—as in “centaur” (Human+AI) chess.

CONCLUSION

Given their rich, complex, and highly-embodied nature, we contend that sports and physical activities represent one of the most promising venues for innovative visualization research today. We identified 21 research opportunities for the visualization and HCI communities, including the need for innovative data-collection methods, sensors, input modalities, displays and interactive hardware that blend with activities, existing equipment, and the constraints of the (sometimes harsh) environments. New technology must be developed based on cross-disciplinary collaborations, participant engagement and co-design, as well as foundational empirical perceptual and cognitive research.

Ultimately, findings and approaches from this research likely have implications that extend beyond sports to a much wider array of possible application areas—where bringing effective situated visualizations to bear on tasks, preserving attention, and preserving social interactions are equally valued.

ACKNOWLEDGMENTS

Light-painting illustrations created by Pearman using Blender atop royalty-free and synthetic base images, with additional refinement via Google Gemini 3.6 Flash by Willett. Perin, Dufresne-Camaro, and Willett co-led the writing of this paper. All authors contributed to editing. Most participated in the initial workshop and subsequent brainstorming and data analysis sessions. Those who did not contributed to visuals and editing.

We thank all the participants who contributed to the IEEE VIS 2024 workshop on First-Person Visualizations for Physical Activities. This work was supported in part by the CRC program and NSERC (RGPIN-2019-05422, RGPIN-2019-05872, RGPIN-2021-02492).

REFERENCES

1. T. Lin, Y. Yang, J. Beyer, and H. Pfister, "Sportsxr–immersive analytics in sports," *arXiv preprint arXiv:2004.08010*, 2020.
2. C. Perin, R. Vuillemot, C. D. Stolper, J. T. Stasko, J. Wood, and S. Carpendale, "State of the Art of Sports Data Visualization," *Computer Graphics Forum*, vol. 37, no. 3, pp. 663–686, 2018.
3. R. Gade, M. Merler, G. Thomas, and T. Moeslund, *The (Computer) Vision of Sports: Recent Trends in Research and Commercial Systems for Sport Analytics*. CRC Press, Jul. 2024, pp. 296–311.
4. A. Li, C. Perin, J. Knibbe, G. Demartini, S. Viller, and M. Cordeil, "Embedded and Situated Visualisation in Mixed Reality to Support Interval Running," *Computer Graphics Forum*, vol. 44, no. 3, p. e70133, 2025.
5. B. Ens, B. Bach, M. Cordeil, U. Engelke, M. Serrano, W. Willett, A. Prouzeau, C. Anthes, W. Büschel, C. Dunne *et al.*, "Grand Challenges in Immersive Analytics," in *Proc. ACM CHI*, 2021.
6. A. Elshabasi, L. Yao, P. Isenberg, C. Perin, and W. Willett, "Collecting Information Needs for Egocentric Visualizations while Running," in *IEEE VIS Workshop on First-Person Visualizations for Outdoor Physical Activities: Challenges and Opportunities*. IEEE, 2024.
7. B. Rey, B. Lee, E. K. Choe, and P. Irani, "Investigating In-Situ Personal Health Data Queries on Smartwatches," *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.*, vol. 6, no. 4, Jan. 2023.
8. C. Perin, T. Lin, L. Yao, Y. Yang, M. Cordeil, and W. Willett, "First-Person Visualizations for Outdoor Physical Activities: Challenges and Opportunities," *Workshops of IEEE VIS*, 2024.
9. T. Blascheck, F. Bentley, E. K. Choe, T. Horak, and P. Isenberg, "Characterizing glanceable visualizations: from perception to behavior change," in *Mobile Data Visualization*. Chapman and Hall/CRC, 2021, pp. 151–176.
10. L. Yao, F. Bucchieri, V. McArthur, A. Bezerianos, and P. Isenberg, "User Experience of Visualizations in Motion: A Case Study and Design Considerations," *IEEE TVCG*, vol. 31, no. 1, pp. 174–184, 2025.

Charles Perin is an Associate Professor of Computer Science at the University of Victoria. Contact him at cperin@uvic.ca.

Charles-Olivier Dufresne-Camaro is a PhD candidate in Computer Science at the University of British Columbia - Okanagan. Contact him at chcamaro@student.ubc.ca.

Maxime Cordeil is an Associate Professor in Data Visualization and Extended Reality at the University of Queensland. Contact him at m.cordeil@uq.edu.au.

Jo Jo Gong is a PhD student in Computer Science at Columbia University. Contact them at ug2155@columbia.edu.

Christophe Hurter is a Professor of Computer Science at the University of Toulouse. Contact him at christophe.hurter@enac.fr.

Pourang P. Irani is a Professor in Computer Science at the University of British Columbia - Okanagan. Contact him at pourang.irani@ubc.ca.

Jade Kandel is a PhD student at the University of North Carolina at Chapel Hill. Contact her at kandelj@cs.unc.edu.

Ang Li is a PhD candidate in Data Visualization and Extended Reality at the University of Queensland. Contact him at angli21@sigchi.org.

Tica Lin is a Senior Research Scientist at Dolby Laboratories in the Advanced Technology Group. Contact her at mlin@g.harvard.edu.

Ben Pearman is a Masters student in Computer Science at the University of Calgary. Contact him at ben.pearman@ucalgary.ca.

Yalong Yang is an Assitant Professor at the School of Interactive Computing, Georgia Tech. Contact him at yalong.yang@gatech.edu.

Lijie Yao is an Assistant Professor of Computer Science at Xi'an Jiaotong-Liverpool University. Contact her at yaolijie0219@gmail.com.

Wesley Willett is an Associate Professor of Computer Science at the University of Calgary. Contact him at wesley.willett@ucalgary.ca.